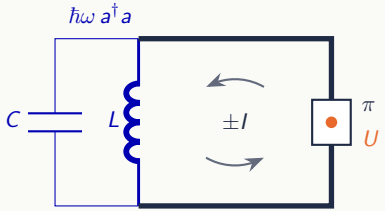
Cavity-induced persistent current in a π loop

Photon condensation, macroscopic quantum tunnelling, and when mean field lies

---

Group seminar

The story in one slide



1. A superconducting loop closed by a π junction shares its inductance with a quantum LC mode.
2. The π junction makes the orbital response paramagnetic $\Rightarrow$ for $\beta > 1$ the mode *softens* and condenses: $\langle a \rangle \neq 0$, a persistent current $\pm I$ flows *without any applied flux*.
3. Quantum fluctuations of the flux *oppose* it: the true ground state is a cat $|\circlearrowleft\rangle + |\circlearrowright\rangle$, split by an instanton gap $\delta \sim \hbar\omega\, e^{-S}$.
4. Full electron–photon DMRG: the effect is real for $E_J \gg \hbar\omega$; the *mean-field* (product-state) prediction is an artefact when $E_J \sim \hbar\omega$.

Outline

1. Motivation: can a cavity make a current?
2. The microscopic model
3. Theory: adiabatic reduction and Landau theory
4. Methods: three levels of description
5. Results
6. Experimental outlook
7. Summary

Motivation: can a cavity make a current?

Photon condensation and the no-go theorem

Question: can coupling matter to a cavity vacuum produce a *static* field / current in the ground state?

- Dicke "superradiant" transition: predicted, then forbidden by the A^2 term.
- **No-go** [Andolina 2019; Nataf 2019]: for a *uniform* cavity field, gauge invariance (f-sum rule) makes the total orbital response diamagnetic $\Rightarrow$ no condensation.
- **Loophole** [Andolina 2020]: a *non-uniform* field (finite $\mathbf{q}$, coupling to $\nabla \times \mathbf{A}$) condenses if the orbital response is paramagnetic and beats the field stiffness.

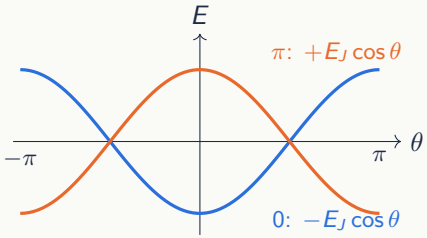
This talk

The minimal realization of the loophole:

- "non-uniform field" $\rightarrow$ flux through one loop;
- "paramagnetic response" $\rightarrow$ a π junction;
- solve it *exactly* (DMRG on electrons + photon), and ask what is quantum and what is classical.

Classical cousin: spontaneous $\Phi_0/2$ currents in π rings [Bulaevskii–Kuzii–Sobyanin 1977] seen in d-wave and SFS loops [Tsuei–Kirtley 2000; Bauer 2004].

Warm-up: 0 and π Josephson junctions



θ = gauge-invariant Cooper-pair phase difference.

- **0 junction:** minimum at $\theta = 0$. Curvature $E'' > 0$: the junction is an *inductance* $L_J = (\Phi_0/2\pi)^2/E_J \Rightarrow$ diamagnetic.
- **π junction:** minimum at $\theta = \pi$; at $\theta = 0$ it sits on a *maximum*: a negative inductance $\Rightarrow$ paramagnetic.
- Realizations: SFS (π from exchange field), d-wave corners, and quantum dots with a local moment (this talk).

Classical π rf-SQUID

$U(\theta) = \frac{E_L}{2}\theta^2 + E_J \cos\theta, \quad E_L = \frac{(\Phi_0/2\pi)^2}{L}.$
$U''(0) = E_L - E_J < 0$ iff $\beta \equiv \frac{E_J}{E_L} = \frac{2\pi L I_c}{\Phi_0} > 1.$

The microscopic model

Step 1: flux enters as a Peierls phase

Tight binding in a vector potential:

$$c_{j+1}^{\dagger} c_j \longrightarrow e^{i\theta_j}\, c_{j+1}^{\dagger} c_j, \qquad \theta_j = \frac{e}{\hbar} \int_j^{j+1} \mathbf{A} \cdot d\mathbf{l}.$$

Gauge transformation $c_j \to e^{i\chi_j} c_j$ moves phase between bonds; only the loop sum is physical:

$$\varphi \equiv \sum_j \theta_j = \frac{e}{\hbar} \oint \mathbf{A} \cdot d\mathbf{l} = \frac{e}{\hbar}\, \Phi.$$

$\Rightarrow$ put the whole φ on *one* bond (next to the junction).

Dictionary (keep this in mind!)

electron phase	$\varphi = e\Phi/\hbar$
Cooper-pair phase	$\theta = 2\varphi$
$\Phi = \Phi_0 = h/2e$	$\varphi = \pi$
$\Phi = \Phi_0/2$	$\varphi = \pi/2$
current	$I = -\frac{e}{\hbar}\, \partial_\varphi E$

With a real pairing Δ the "one-bond" gauge is *not* the same as a uniform $\mathbf{A}$; it equals uniform $\mathbf{A}$ with covariant $\Delta_j \to \Delta e^{-2i\chi_j}$ (checked to 10^{-10}; backup).

Step 2: quantize the flux of the LC mode

$$H_{LC} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} = \hbar\omega\left(a^\dagger a + \tfrac{1}{2}\right)$$

$$\Phi = \Phi_{\mathrm{zpf}}\left(a + a^\dagger\right), \qquad \Phi_{\mathrm{zpf}} = \sqrt{\hbar Z/2}$$

($Z = \sqrt{L/C}$). The loop phase becomes an operator:

$$\varphi \;\to\; \varphi_{\mathrm{cl}} + g\left(a + a^\dagger\right), \qquad g = \frac{e\,\Phi_{\mathrm{zpf}}}{\hbar}$$

g in circuit language

$g^2 = \dfrac{e^2 Z}{2\hbar} = \pi\dfrac{e^2}{h}Z \;\Rightarrow\; (2g)^2 = \pi\dfrac{Z}{R_Q}$,

$R_Q = h/4e^2 \simeq 6.45\,\mathrm{k\Omega}$.

$Z = 50\,\Omega \Rightarrow g \approx 0.08$;

$Z \approx 120\,\Omega \Rightarrow g \approx 0.12$.

The junction hopping becomes a photon displacement:

$$-t_J\, e^{i\varphi_{\mathrm{cl}}}\, \underbrace{e^{ig(a+a^\dagger)}}_{D(ig)}\, c^\dagger_{1\sigma} c_{0\sigma} + \mathrm{h.c.}$$

- each tunnelling electron kicks the oscillator (displacement D);
- expanding: $e^{igX} = 1 + igX \underbrace{-\frac{g^2}{2}X^2}_{A^2\text{ term}} + \ldots$
  $\Rightarrow$ diamagnetic terms included to all orders;
- manifestly gauge invariant: no A^2 "bookkeeping" errors possible.

The full Hamiltonian

$$H = \hbar\omega\, a^\dagger a \underbrace{- t \sum_{r=1}^{L-2} \sum_{\sigma} \left(c^\dagger_{r+1\sigma} c_{r\sigma} + \text{h.c.} \right) + \Delta \sum_{r=1}^{L-1} \left(c^\dagger_{r\uparrow} c^\dagger_{r\downarrow} + \text{h.c.} \right)}_{s\text{-wave BCS chain (the “leads”)}}$$

$$+ \underbrace{\epsilon_d\, n_0 + U\, n_{0\uparrow} n_{0\downarrow}}_{\text{Anderson dot, site 0}} - t_J \sum_{\sigma} \left(c^\dagger_{0\sigma} c_{L-1\,\sigma} + \text{h.c.} \right)$$

$$- t_J \sum_{\sigma} \left(e^{i\varphi_{\text{cl}}}\, e^{ig(a+a^\dagger)}\, c^\dagger_{1\sigma} c_{0\sigma} + \text{h.c.} \right) \equiv \hbar\omega\, a^\dagger a + H_{\text{ring}}\left[\varphi_{\text{cl}} + g(a + a^\dagger)\right]$$

Default parameters

$t = 1$, $\Delta = t$, $t_J = 0.5t$, $L = 14$ ($\gg \xi \simeq 2t/\Delta$),
$\hbar\omega = 7.6 \times 10^{-4} t$,
$\epsilon_d = -U/2$ (p–h symmetric); no external flux:
$\varphi_{\text{cl}} = 10^{-3}$ seed only.

Auxiliary versions: (i) weak link ($U = 0$) biased by $\Phi_0/2$; (ii) short ring with a Zeeman-polarized dot (the original mean-field notebook).

All the ring physics is in one function: $E(\varphi)$

Ground-state energy of H_{ring} at fixed (classical) phase:

$$E(\varphi) = \sum_{m \geq 0} E_m \cos m\varphi$$

- **Only cosines:** $H_{\text{ring}}[\varphi]^* = H_{\text{ring}}[-\varphi]$ (time reversal) $\Rightarrow E(\varphi) = E(-\varphi)$.
- $m = 2, 4, \ldots$: charge $2e$ crossing $\Rightarrow$ $h/2e$ periodicity (Josephson).
- $m = 1$: a single electron encircling the ring [Loss–Goldbart 1991], $\propto e^{-L/\xi}$: a mesoscopic nuisance.
- Define $E_J \equiv |E_2|$, $I_c = 2eE_J/\hbar$.

Sign convention

	E_2	$E''(0)$
0 junction	< 0	> 0 (diamagnetic)
π junction	> 0	< 0 (paramagnetic)

For a pure π junction, $E(\varphi) = E_J \cos 2\varphi$:

$$I(\varphi) = -\frac{e}{\hbar} E'(\varphi) = \underbrace{\frac{2eE_J}{\hbar}}_{I_c} \sin 2\varphi.$$

Two routes to a π loop — and why they differ

Route A: 0 junction + external $\Phi_0/2$ ($\varphi_{\rm cl} = \pi/2$):

$$E_2 \cos 2(\varphi + \tfrac{\pi}{2}) = -E_2 \cos 2\varphi \quad \checkmark$$
$$E_1 \cos(\varphi + \tfrac{\pi}{2}) = -E_1 \sin\varphi \quad \text{(odd!)}$$

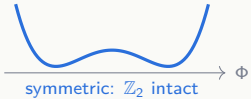
The h/e harmonic becomes a tilt of the double well: it *selects* the current direction. Also needs a flux stable to $|\delta\Phi| \ll \delta/2I^\star$.



Route B: intrinsic π junction ($\varphi_{\rm cl} = 0$):

- $H_{\rm ring}[0]$ is real $\Rightarrow E(\varphi) = E(-\varphi)$ *exactly*.
- The two current directions are related by time reversal: a genuine $\mathbb{Z}_2$ symmetry, spontaneously broken (or tunnelling-restored).
- Our choice: an interacting dot, $U > U_c$.

Up to odd harmonics the two routes give the *same* $E(\varphi)$ (shown later, DMRG).



Theory: adiabatic reduction and Landau theory

Born–Oppenheimer: electrons are fast, the flux is slow

Scales: $\hbar\omega \sim 10^{-3}t \ll E_A$ (Andreev gap ~ 0.1–$1\,t$).

Introduce flux quadratures:

$$X = a + a^\dagger, \quad P = i(a^\dagger - a), \quad [X, P] = 2i$$

$$\hbar\omega\, a^\dagger a = \tfrac{\hbar\omega}{4}\left(X^2 + P^2\right) - \tfrac{\hbar\omega}{2}.$$

Ansatz $|\Psi\rangle = \int dX\, \chi(X)\, |X\rangle \otimes |\psi_0(\varphi_{\rm cl} + gX)\rangle$ with $H_{\rm ring}[\varphi]|\psi_0(\varphi)\rangle = E(\varphi)|\psi_0(\varphi)\rangle$:

$$H_{\rm BO} = \frac{\hbar\omega}{4}\left(P^2 + X^2\right) + E(\varphi_{\rm cl} + gX)$$

- A 1D Schrödinger problem for the flux; $P = -2i\,\partial_X$.
- The photon is treated exactly (all Fock states, all correlations); only adiabaticity is assumed.
- Neglected: diagonal BO term $\propto \hbar\omega\, g^2 \langle \partial_\varphi \psi_0 | \partial_\varphi \psi_0 \rangle$, relative size $\sim \hbar\omega / E_A$.
- Uses the *full* $E(\varphi)$, not only the first harmonic.
- Checked against full DMRG: $\lesssim$ few % everywhere.

It is an rf-SQUID (fluxonium) whose inductance is the mode

Change variables to the Cooper-pair phase and its conjugate charge:

$$\theta = 2gX, \qquad n = \frac{P}{4g}, \qquad [\theta, n] = i.$$

Two lines of algebra

$$\frac{\hbar\omega}{4}X^2 = \frac{\hbar\omega}{16g^2}\theta^2 \equiv \frac{1}{2}E_L\theta^2, \quad E_L = \frac{\hbar\omega}{8g^2}$$

$$\frac{\hbar\omega}{4}P^2 = 4\hbar\omega g^2\, n^2 \equiv 4E_C n^2, \quad E_C = \hbar\omega\, g^2$$

Consistency: $\sqrt{8E_C E_L} = \hbar\omega$ ✓

$$H_{\mathrm{BO}} = 4E_C\, n^2 + \tfrac{1}{2}E_L\,\theta^2 + E(\theta/2)$$

- Pure π junction: $E(\theta/2) = +E_J\cos\theta$ $\Rightarrow$ a π-fluxonium.
- g controls the *ratio* $E_L/E_C = 1/(8g^4)$: small g = heavy, classical flux.
- Two dimensionless knobs:

$$\beta = \frac{E_J}{E_L}, \qquad \frac{E_J}{\hbar\omega}.$$

Landau theory: when does the mode condense?

Classical energy (drop P):

$$\mathcal{F}(X) = \frac{\hbar\omega}{4}X^2 + E(gX)$$

$$\mathcal{F}''(0) = \frac{\hbar\omega}{2} + g^2E''(0) = \frac{\hbar\omega}{2}(1-\beta)$$

Instability criterion

$$\beta \equiv \frac{2g^2|E''(0)|}{\hbar\omega} > 1 \quad \text{with} \quad E''(0) < 0$$

Pure π junction, $E''(0) = -4E_J$:

$$\beta = \frac{8g^2E_J}{\hbar\omega} = \frac{E_J}{E_L} = \frac{2\pi LI_c}{\Phi_0}$$

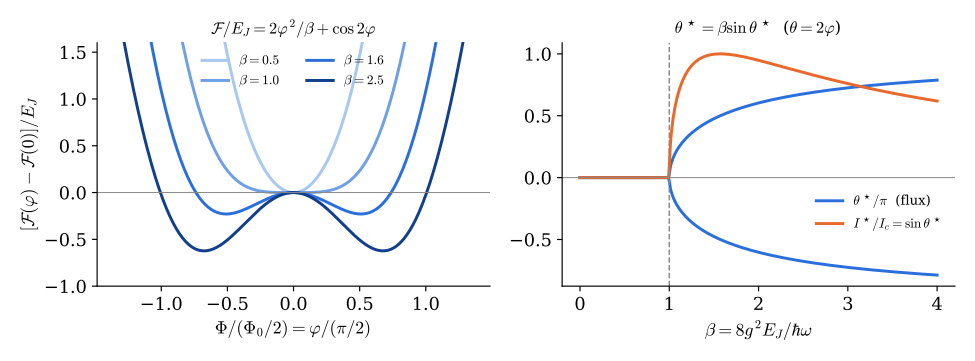
$=$ the textbook screening parameter of the loop.

- Needs paramagnetic response ($E'' < 0$): a 0 junction only *stiffens* the mode.
- This is the single-loop version of the photon-condensation criterion of Andolina *et al.*
- Why the no-go theorem does not apply: the mode couples to the *flux* (curl of **A**), and the π shift supplies the paramagnetism.

Numbers ($U = 2t$ dot)

$E_J = 7.39 \times 10^{-3}t$, $\hbar\omega = 7.6 \times 10^{-4}t$
$\Rightarrow E_J/\hbar\omega \simeq 9.7$
$\beta = 1 \Rightarrow g_c^{\mathrm{cl}} = \sqrt{\hbar\omega/8E_J} = 0.113$
DMRG: $g_c \simeq 0.12$ ✓

The classical double well and the pitchfork



$\mathcal{F}'(X) = 0$ with $\theta = 2gX$:

$$\frac{\hbar\omega}{2g}\,\varphi = 2gE_J \sin 2\varphi \;\Rightarrow\; \boxed{\theta^\star = \beta \sin\theta^\star}$$

$I^\star = I_c \sin\theta^\star$. Large β: $\theta^\star \to \pi$ (flux $\to \Phi_0/2$) and $I^\star \simeq \pi I_c/\beta \propto \hbar\omega/g^2$ *decreases*.

An exact identity: the current *is* the displacement of the mode

In any eigenstate $\langle [H, P] \rangle = 0$. Compute the commutator:

$$[a^\dagger a, P] = i\,(a^\dagger + a) = iX,$$
$$[H_{\mathrm{ring}}(\varphi_{\mathrm{cl}} + gX), P] = 2i\,g\,\partial_\varphi H_{\mathrm{ring}}.$$

With $I \equiv -\frac{e}{\hbar}\partial_\varphi H_{\mathrm{ring}}$:

$$\langle I \rangle = \frac{e}{\hbar}\,\frac{\hbar\omega\,\langle X \rangle}{2g} \qquad \Longleftrightarrow \qquad \langle I \rangle = \frac{\langle \Phi \rangle}{L}$$

(use $2\Phi_{\mathrm{zpf}}^2 = \hbar\omega L$). Kirchhoff's law, but exact for the full quantum problem.

Why it matters

- Current flows $\Leftrightarrow$ the field has a static part, $\langle a \rangle \neq 0$: a photon condensate.
- Condensate size: $\langle a^\dagger a \rangle \gtrsim \langle X \rangle^2/4 = (gI\hbar/e\hbar\omega)^2$. $U = 2t$, $g = 0.14$: ≈ 6.5 photons (DMRG: ≈ 8).
- Numerically: $\langle X \rangle$ converges much better in DMRG than the bond current (a small difference of large terms) $\Rightarrow$ we measure I through X.
- Verified in ED to 10^{-7}.

Below threshold: the soft mode

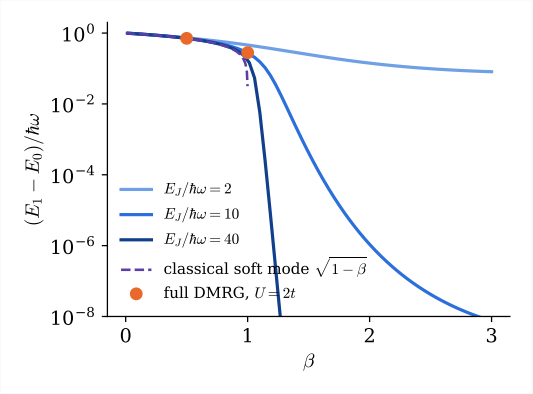
Expand to second order around $X = 0$ ($\beta < 1$):

$$H_{\mathrm{BO}} \simeq \frac{\hbar\omega}{4}\Big(P^2 + (1-\beta)X^2\Big)$$

- Excitation gap: $\boxed{\hbar\omega\sqrt{1-\beta}}$ (mode softening).
- Vacuum flux is squeezed: $\langle X^2\rangle = (1-\beta)^{-1/2}$.
- Photons in the ground state: $\langle n\rangle = \frac{1}{4}\left[(1-\beta)^{-\frac{1}{2}} + (1-\beta)^{\frac{1}{2}}\right] - \frac{1}{2}$.

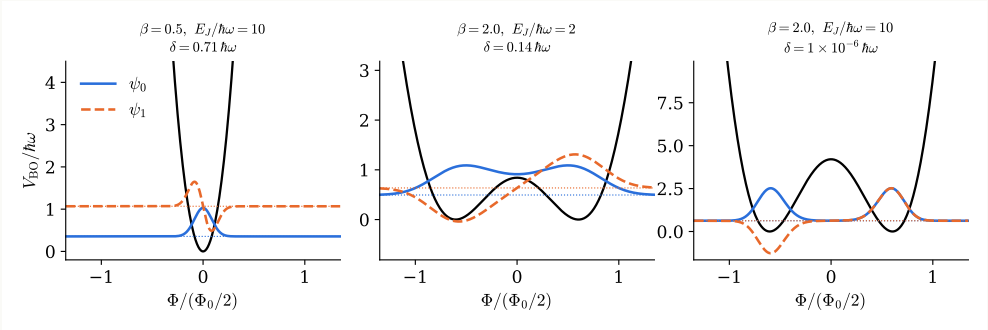
Check against full DMRG

$\beta = 0.5$: $\sqrt{0.5} = 0.707$ vs DMRG 0.71
$\beta = 0.57$: $\langle n\rangle = 0.045$ vs DMRG 0.04



At $\beta \to 1$ the quadratic theory fails (quartic oscillator); above it, the gap becomes the tunnel splitting and collapses exponentially.

Quantum fluctuations oppose the transition



Exact eigenstates of H_{BO} (pure π junction). For $\beta > 1$: $\psi_{0,1} \simeq (|\circlearrowleft\rangle \pm |\circlearrowright\rangle)/\sqrt{2}$ — cat states of macroscopic current, split by δ.

Zero-point fluctuations *restore* the symmetry: if the barrier is below $\sim \hbar\omega$ (middle) there is no current-carrying state at all.

How small is δ? An instanton estimate

Particle of mass $M = \hbar^2/8E_C$ in $U(\theta) = E_J\, u(\theta)$, $u = \frac{\theta^2}{2\beta} + \cos\theta$. WKB:

$$\delta \sim \hbar\omega\, e^{-S}, \quad S = \int_{-\theta^\star}^{\theta^\star} d\theta \sqrt{\frac{U - U_{\min}}{4E_C}}$$

Scaling

$$\frac{E_J}{4E_C} = \frac{E_J}{4g^2\hbar\omega} = \frac{2}{\beta}\left(\frac{E_J}{\hbar\omega}\right)^2 \Rightarrow \boxed{S = \frac{E_J}{\hbar\omega}\, s(\beta)}$$

$s(\beta) = \sqrt{2/\beta}\int d\theta\, \sqrt{u - u_{\min}}$.

Near threshold ($\epsilon = 1 - 1/\beta$): $u \simeq 1 - \frac{\epsilon}{2}\theta^2 + \frac{\theta^4}{24}$, minima at $\theta^2 = 6\epsilon$, depth $\frac{3}{2}\epsilon^2 \Rightarrow s \propto (\beta - 1)^{3/2}$.

Sharp transition only in a scaling limit

$E_J/\hbar\omega \to \infty$ at fixed β:

$$\frac{E_C}{E_J} = \frac{\beta}{8}\left(\frac{\hbar\omega}{E_J}\right)^2 \to 0,$$

flux becomes classical, $\delta \to 0$, pitchfork with mean-field exponents.

Cf. quantum Rabi model: QPT only for $\Omega/\omega \to \infty$ [Hwang–Puebla–Plenio 2015]. Finite $E_J/\hbar\omega$: a crossover; δ = macroscopic quantum tunnelling of the current.

What the cavity does and does not do

Static threshold is set by the loop inductance.

Loop with its own L_r, mutual $M \leq \sqrt{L_r L_c}$ to the resonator: minimizing the static energy over the resonator flux leaves $\Phi_r^2/2L_r$.

- A resonator cannot push the static threshold below what the loop's own inductance gives.
- Our model is the galvanic limit $k = M/\sqrt{L_r L_c} \to 1$ (resonator inductance *is* the loop inductance, as in fluxonium).

Honest framing

- The instability is *classical*: a static field, $\beta > 1$.
- Vacuum fluctuations do not drive it, they oppose it (tunnelling δ).
- The mode supplies the *quantum dynamics* of the flux (E_C), the condensate's photons, and a readout channel.

Methods: three levels of description

Three ways to solve $H = \hbar\omega a^\dagger a + H_{\mathrm{ring}}[\varphi_{\mathrm{cl}} + gX]$

	ansatz	photon	cost
Mean field (MF)	$\lvert\psi_{\mathrm{el}}\rangle \otimes \lvert\chi_{\mathrm{ph}}\rangle$, self-consistent (HF–BdG for U)	product state	trivial
Adiabatic (BO)	$\chi(X)\,\lvert\psi_0(gX)\rangle$, $E(\varphi)$ from BdG or DMRG	exact	1D Schrödinger
Full DMRG	MPS on electrons $\otimes$ photon, no approximation	exact	$\sim$h on 4 cores

Strategy: DMRG is the referee. Once BO is validated against DMRG, use BO to map whole phase diagrams and to reach $\delta \ll$ DMRG resolution.

Mean field and its two hidden assumptions

Product state $\Rightarrow$ two coupled problems:

$$\text{el.: } H_{\text{ring}} \text{ with } t_J \to t_J\,\alpha, \quad \alpha = \langle\chi|e^{igX}|\chi\rangle$$

$$\text{ph.: } \hbar\omega a^\dagger a - \left(\eta\, e^{igX} + \text{h.c.}\right),$$

$$\eta = t_J e^{i\varphi_{\text{cl}}} \textstyle\sum_\sigma \langle c_{1\sigma}^\dagger c_{0\sigma}\rangle$$

$$E_{\text{MF}} = E_{\text{el}} + E_{\text{ph}} + 2\,\text{Re}(\eta\alpha) \quad \text{(double counting)}$$

For a coherent state $|X_0\rangle$: $\alpha = e^{igX_0}\, e^{-g^2/2}$ (Debye–Waller).

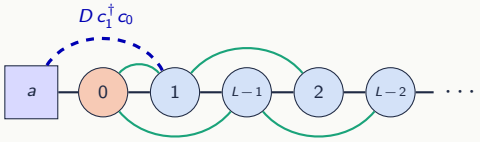
(i) Not gauge invariant

All phase on one bond: that bond is shrunk by $e^{-g^2/2}$ and loses *phase-independent* kinetic energy. Spread over N bonds: loss $\sim N \cdot g^2/2N^2 \to 0$. The exact state recovers it by following the flux adiabatically. [Li *et al.* 2020; Dmytruk–Schirò 2021]

(ii) Self-trapping

A product state must pick one well: it breaks the $\mathbb{Z}_2$ by construction, even when the barrier is below the zero-point energy.

DMRG for electrons + photon



ring bonds: range ≤ 2 (folded order)

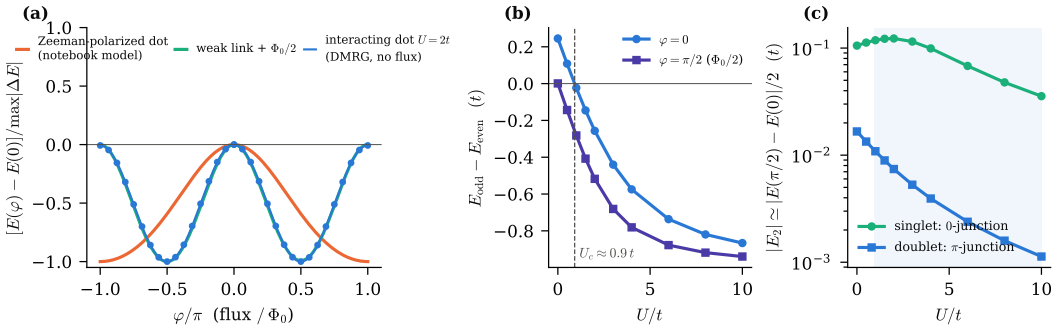
- TeNPy; photon = bosonic site, Fock cutoff $N_{\mathrm{ph}} = 30$–50 (discarded weight $< 10^{-7}$).
- Conserved: fermion parity, S^z (not N: Δ breaks U(1)).
- Vertex $e^{igX} c_{1\sigma}^\dagger c_{0\sigma}$ = three-site MPO term.
- $\chi = 256$–400; E_0 error $\lesssim 3 \times 10^{-6} t$.
- Excited states: DMRG orthogonal to $|\Psi_0\rangle$.

Benchmarks

$g = 0$ vs exact BdG: 10^{-14} ($\chi = 1024$) · 4-site ring + photon vs ED: 10^{-15} · $\langle I \rangle = \hbar\omega \langle X \rangle / 2g$ to 10^{-7}

Results

The junction: a π shift from Coulomb repulsion



(a) After normalization, a $U = 2t$ dot at *zero* flux $\equiv$ weak link + $\Phi_0/2$. The short-ring notebook model is dominated by the h/e harmonic.

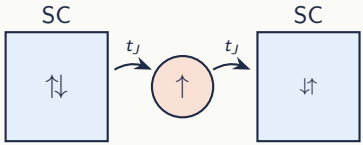
(b) Parity crossing at $U_c \simeq 0.9t$: for $U > U_c$ the doublet (local moment) is the ground state at all φ.

(c) $E_2(U = 2t) = +7.39 \times 10^{-3} t$, $|E_1|, |E_4| < 10^{-4} t$; $E_2 \propto U^{-1.3}$ (cotunnelling).

Why a local moment gives a π junction

Singly occupied dot ($U \gg t_J, \Delta$): a Cooper pair crosses by *fourth-order cotunnelling*, one electron at a time, through virtual states.

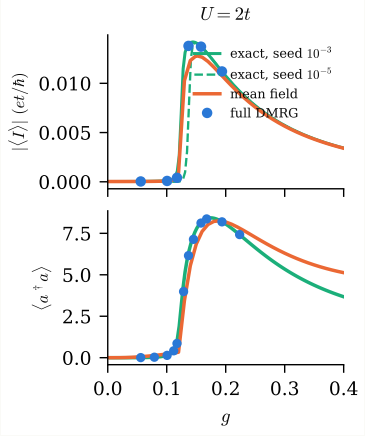
- The dot's own electron must leave and be replaced; the pair arrives with its two spins in reversed order.
- Singlet: $c_\downarrow^\dagger c_\uparrow^\dagger = -c_\uparrow^\dagger c_\downarrow^\dagger \Rightarrow$ the pair-transfer amplitude changes sign: $E_J \to -E_J$.
- [Glazman–Matveev 1989; Spivak–Kivelson 1991]; seen in nanowire and CNT dots [van Dam 2006; Cleuziou 2006].



4 hops, intermediate energies $\sim U, \Delta$

Bonus: the low-energy manifold is *fourfold*, $\{\uparrow,\downarrow\}_{\mathrm{dot}} \otimes \{\circlearrowleft,\circlearrowright\}$, split only by δ.

Central result: spontaneous current at $U = 2t$, no applied flux

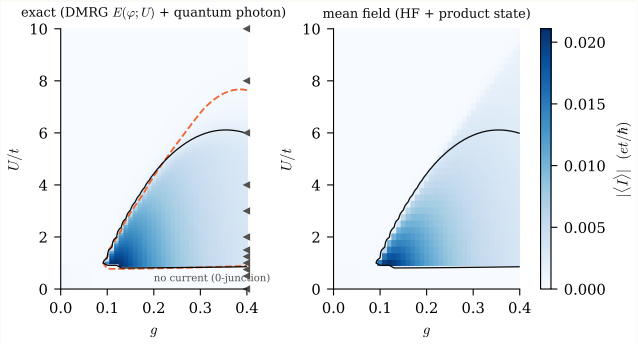


- $g < g_c \simeq 0.12$: mode near vacuum, $|\langle I \rangle| \lesssim 10^{-4}\, et/\hbar =$ linear response to the seed.
- $g \simeq g_c$: photon number jumps to $\simeq 8$, current $1.4 \times 10^{-2}\, et/\hbar$.
- **Spontaneous:** seed $10^{-3} \to 10^{-5}$: current above threshold unchanged, below threshold $\propto$ seed.
- Full DMRG (dots) and exact-photon BO (lines) agree within a few %.

Sanity check with the formulas

$I_c = 2E_J = 0.0148\, et/\hbar \Rightarrow I_{\max} \simeq 0.93\, I_c$.
Tail: $\langle I \rangle = \hbar\omega\varphi^\star/2g^2$; at $g = 0.4$, $\varphi^\star \to \pi/2$:
$7.6 \cdot 10^{-4} \cdot 1.57/0.32 = 3.7 \times 10^{-3}$ ✓

The (U, g) phase diagram: exact vs mean field



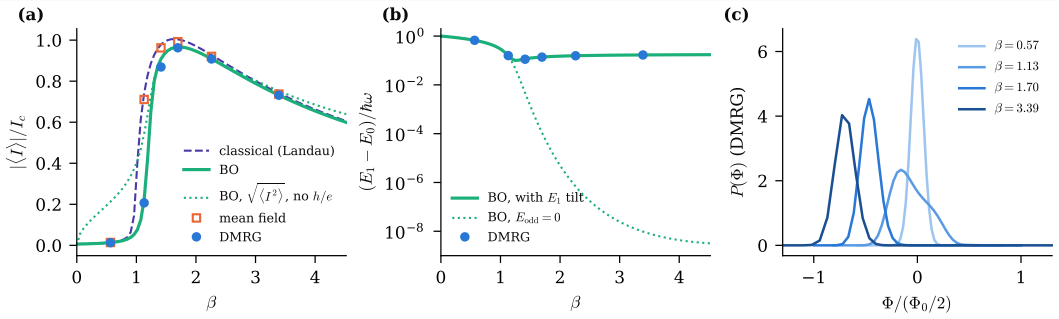
Black: $|\langle I \rangle| = 2 \times 10^{-3}\, et/\hbar$ (exact); orange dashed: same contour from MF; arrows: U values computed by DMRG.

- Lower edge $U_c \simeq 0.9t$: the 0–π transition of the dot. No current for $U < U_c$ at *any* g.
- Left edge $g_c(U)$: follows $\beta(U, g) = 1$; moves right as $E_2(U)$ drops.

U/t	g_c ex.	g_c MF	$I_{\max}$ ex.	MF
2	0.13	0.13	0.0141	0.0128
4	0.19	0.19	0.0070	0.0057
6	0.32	0.27	0.0023	0.0031
7	none	0.32	$6{\cdot}10^{-4}$	0.0024
8	none	0.4	$2{\cdot}10^{-4}$	0.0019

MF fine for $E_J \gtrsim 3\hbar\omega$; for $U \gtrsim 6t$ it invents an ordered phase.

Route A: weak link biased by $\Phi_0/2$ ($E_J/\hbar\omega = 10$)

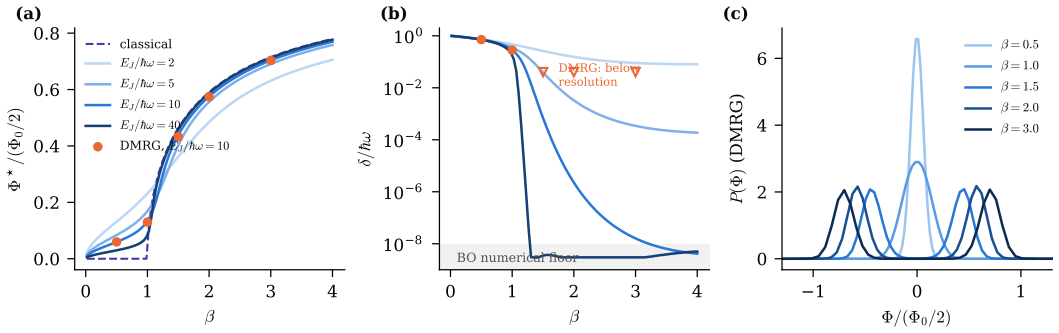


(a) Current rises at $\beta \simeq 1.1$, peaks at $0.97 I_c$. MF: 3× too large near threshold (0.71 vs 0.21 I_c at $\beta = 1.13$), 0.2–0.4$\hbar\omega$ too high in energy.

(b) Splitting saturates at $\sim 0.15\hbar\omega$: it is the tilt $2|E_1| \sin\varphi^\star \approx 0.2\hbar\omega$, not tunnelling. Without E_1: $\delta < 10^{-6}\hbar\omega$ at $\beta = 2$.

(c) $P(\Phi)$ from the photon reduced density matrix: a *single* displaced peak — direction *selected*, not spontaneous.

Route B: interaction-driven π loop ($U = 2t$, $\varphi_{\mathrm{cl}} = 0$)

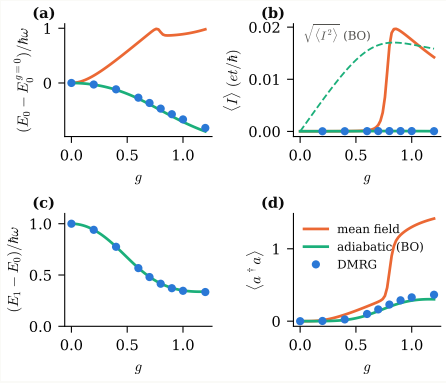


(a) $\langle I \rangle = 0$ in every eigenstate (TRS)! Order parameter: eigenvalues $\pm\Phi^\star$ of the flux projected on the lowest doublet. DMRG vs BO within 3%.

(b) $\delta/\hbar\omega = 0.71, 0.28$ at $\beta = 0.5, 1$ (both methods); then BO: $7 \cdot 10^{-4}$, $2 \cdot 10^{-6}$, $2 \cdot 10^{-8}$ at $\beta = 1.5, 2, 3$.

(c) DMRG ground state is a cat: symmetric, bimodal $P(\Phi)$ with peaks at $\pm\Phi^\star$ tracking the classical minimum.

Cautionary tale: the original mean-field prediction



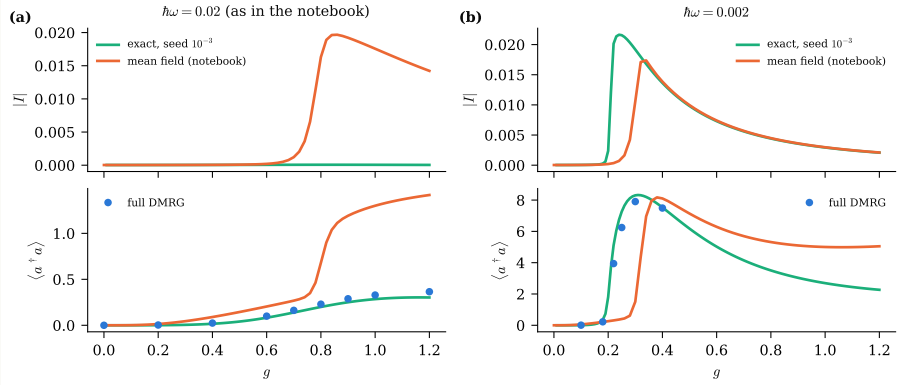
Short ring, Zeeman-polarized dot, $\hbar\omega = 0.02$:

- MF: current-carrying phase for $g \gtrsim 0.75$, $\langle n \rangle \simeq 1.4$.
- DMRG: no current (only linear response, 7×10^{-5}), energy $1.8\hbar\omega$ *below* MF, gap stays $0.33\hbar\omega$, $\langle n \rangle \simeq 0.36$.
- BO $\approx$ DMRG to $0.06\hbar\omega$ in energy and 1.3% in gap.

Three reasons

(1) gauge non-invariance of the product state; (2) $E_J \simeq \hbar\omega$: double well shallower than zero-point energy; (3) $L \lesssim \xi$:
$E_1 = 2.1 \times 10^{-2} \gg E_2 = 2.0 \times 10^{-3}$ — the instability would be of the *normal* h/e current, not Josephson.

Same model, ten times smaller $\hbar\omega$: the effect comes back



At $\hbar\omega = 0.02$ the exact ground state carries no current at any g. At $\hbar\omega = 0.002$ the same model orders: exact $g_c \simeq 0.21$, MF $g_c \simeq 0.30$. The control parameter is $E_J/\hbar\omega$, not g alone.

When can we trust the product state?

MF works

- $E_J \gg \hbar\omega$ *and* $g \ll 1$;
- deep in the ordered phase (photon $\approx$ coherent state);
- gets U_c and $g_c(U)$ right for $U \lesssim 5t$; current low by 10–20% (bond renormalized by $|\langle D \rangle| < 1$).

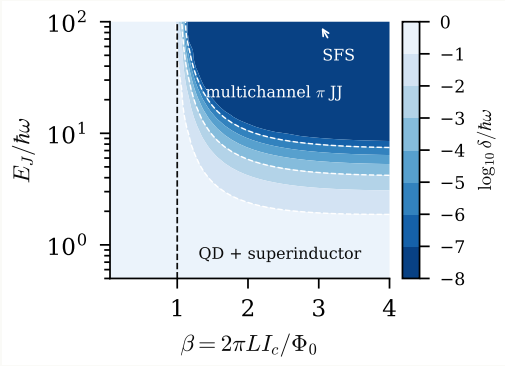
MF fails

- near threshold (overestimates current $\times 3$);
- $E_J \lesssim 3\hbar\omega$: predicts order where the exact state is a symmetric cat;
- always: energy too high by 0.2–$2\hbar\omega$; misses δ and cat structure entirely.

Cheap and safe alternative: **BO with exact photon** — one ring calculation $E(\varphi)$ plus a 1D Schrödinger equation. Agrees with full DMRG to a few % throughout.

Experimental outlook

Where are real devices on the map?



δ from BO with the DMRG $E(\varphi)$, $U = 2t$.

$$\beta = 2\pi \frac{Z}{R_Q}\frac{E_J}{\hbar\omega}, \qquad E_J/h \simeq 0.5\,\mathrm{GHz} \times I_c[\mathrm{nA}]$$

- **SFS** ($I_c \sim$ 1–100 μA): $E_J/\hbar\omega \sim 10^2$–$10^4$. Classical limit; spontaneous currents already observed.
- **QD π junctions** ($I_c \sim$ 0.1–10 nA): need superinductances (0.03–3 μH); $E_J/\hbar\omega \sim 1$, cat with $\delta \sim 0.1\hbar\omega$ — the π-qubit regime. Signature: dip of the $0 \to 1$ frequency, jump of $\langle 0|\Phi|1\rangle$ (two-tone spectroscopy).
- **Sweet spot**: multichannel / gate-tunable π JJ, $I_c \sim$ 50–200 nA, $L \sim$ 2–10 nH, $\omega/2\pi \sim$ 2–5 GHz: $\beta \sim$ 1–3, $E_J/\hbar\omega \sim$ 10–30, $\delta/\hbar\omega \sim 10^{-4}$–$10^{-8}$.

Summary

Take-home messages

1. A π junction in a loop is the minimal photon condensate: paramagnetic orbital response + flux coupling evade the no-go theorem. Criterion: $\beta = 2g^2|E''(0)|/\hbar\omega > 1$.
2. Everything reduces to a π-fluxonium $4E_C n^2 + \frac{1}{2}E_L\theta^2 + E(\theta/2)$ with $E_C = g^2\hbar\omega$, $E_L = \hbar\omega/8g^2$; exact identity $\langle I\rangle = \langle\Phi\rangle/L$.
3. The transition is classical in origin; quantum fluctuations restore symmetry via a cat state with $\delta \sim \hbar\omega\, e^{-(E_J/\hbar\omega)s(\beta)}$. Sharp only for $E_J/\hbar\omega \to \infty$.
4. An *interacting* dot ($U > U_c \simeq 0.9t$) gives a TRS-protected π loop: current direction chosen spontaneously, no flux bias to stabilize; fourfold spin$\otimes$current multiplet.
5. Full electron–photon DMRG confirms it for $E_J \gg \hbar\omega$ and shows that MF invents the effect when $E_J \sim \hbar\omega$. BO with an exact photon is the reliable cheap tool.

Open questions

- **Kondo regime** ($T_K \gg \Delta$): the junction returns to 0 — how does the condensate disappear?
- **Self-consistent leads**: with a rigid order parameter all phase drops on the junction; a self-consistent BdG ring puts most of it in the leads (backup). How does lead kinetic inductance renormalize β?
- **Spin $\otimes$ current multiplet**: can the $\{\uparrow,\downarrow\} \otimes \{\circlearrowright,\circlearrowleft\}$ quartet be used/read out?
- **Dissipation and readout**: what does a real resonator measure — $\langle 0|\Phi|1\rangle$, dispersive shift?
- **Beyond one mode**: multimode / transmission-line resonators, finite $k < 1$.

Thank you!

Backup

Backup: gauge covariance of the Peierls placement

Non-interacting ring with a dot ($L = 14$, $\Delta = t$, $t_J = 0.5t$); BdG ground-state energy:

φ	all phase on one bond	uniform $\mathbf{A}$ + covariant Δ_j	uniform $\mathbf{A}$ + real Δ
0	-21.82094	-21.82094	-21.82094
0.5	-21.79739	-21.79739	-21.81251
1.0	-21.72957	-21.72957	-21.78725
$\pi/2$	-21.61009	-21.61009	-21.73791

First two columns agree to 10^{-10} (`code/tests/test_gauge_covariance.py`). A uniform $\mathbf{A}$ with a *real* Δ is a different (current-non-conserving) model — compare Dmytruk & Schirò.

Backup: where does the phase drop? (self-consistent BdG)

Model assumption: fixed real $\Delta \Rightarrow$ rigid phase in the leads (zero lead kinetic inductance), *all* phase drops on the junction.

Check: weak link ($t_J = 0.5$, $U = 0$, $L = 14$) with $\Delta_j = -U_a \langle c_{j\downarrow} c_{j\uparrow} \rangle$ solved self-consistently (complex), $U_a = 3.07$ tuned to $|\Delta| = 1$:

pair flux 2φ	junction drop	fraction
0.4	0.110	0.275
0.8	0.221	0.277
1.2	0.335	0.279

- Current is conserved bond by bond (as it must be self-consistently).
- In this short ring only $\approx 28\%$ of the gauge-invariant phase drops on the junction: the 13 lead bonds act as a series kinetic inductance.
- In lumped language this adds to the loop inductance; for wide/macroscopic leads it is small. Worth quantifying for the paper.

code/selfconsistent_phase.py

Backup: DMRG numbers

Weak link $+\Phi_0/2$, $E_J/\hbar\omega = 10$, $\chi = 256$, $N_{\mathrm{ph}} = 40$

β	I/I_c DMRG	BO	MF	$\delta/\hbar\omega$	$\langle n \rangle$
0.57	0.013	0.014	0.014	0.670	0.04
1.13	0.207	0.223	0.711	0.159	1.2
1.41	0.869	0.908	0.962	0.113	5.2
1.70	0.963	0.967	0.993	0.138	7.0
2.26	0.908	0.909	0.921	0.156	8.3
3.39	0.732	0.732	0.737	0.168	8.1

$U = 2t$ **dot**, $\varphi_{\mathrm{cl}} = 0$, $E_J/\hbar\omega = 10$, $\chi = 256$, $N_{\mathrm{ph}} = 50$

β	$\Phi^*/(\Phi_0/2)$ DMRG / BO	$\delta/\hbar\omega$ DMRG / BO
0.5	0.060 / 0.060	0.71 / 0.71
1.0	0.130 / 0.134	0.28 / 0.28
1.5	0.433 / 0.438	(< 0.04) / $7 \cdot 10^{-4}$
2.0	0.574 / 0.575	(< 0.04) / $2 \cdot 10^{-6}$
3.0	0.703 / 0.704	(< 0.04) / $2 \cdot 10^{-8}$

DMRG excited-state resolution $\sim 3 \times 10^{-5} t \approx 0.04\hbar\omega$ (from $\chi = 256$ vs 400).

Backup: references

- L. N. Bulaevskii, V. V. Kuzii, A. A. Sobyanin, JETP Lett. **25**, 290 (1977)
- C. C. Tsuei, J. R. Kirtley, RMP **72**, 969 (2000); V. V. Ryazanov *et al.*, PRL **86**, 2427 (2001); A. Bauer *et al.*, PRL **92**, 217001 (2004)
- G. M. Andolina *et al.*, PRB **100**, 121109(R) (2019); PRB **102**, 125137 (2020); P. Nataf *et al.*, PRL **123**, 207402 (2019)
- M.-J. Hwang, R. Puebla, M. B. Plenio, PRL **115**, 180404 (2015)
- L. I. Glazman, K. A. Matveev, JETP Lett. **49**, 659 (1989); B. I. Spivak, S. A. Kivelson, PRB **43**, 3740 (1991)
- E. Vecino *et al.*, PRB **68**, 035105 (2003); C. Karrasch *et al.*, PRB **77**, 024517 (2008)
- J. A. van Dam *et al.*, Nature **442**, 667 (2006); J.-P. Cleuziou *et al.*, Nat. Nano. **1**, 53 (2006); R. Delagrange *et al.*, PRB **91**, 241401(R) (2015)
- L. B. Ioffe *et al.*, Nature **398**, 679 (1999); T. Yamashita *et al.*, PRL **95**, 097001 (2005); A. K. Feofanov *et al.*, Nat. Phys. **6**, 593 (2010)
- J. R. Friedman *et al.*, Nature **406**, 43 (2000); V. E. Manucharyan *et al.*, Science **326**, 113 (2009); L. Grünhaupt *et al.*, Nat. Mater. **18**, 816 (2019)
- J. Li *et al.*, PRB **101**, 205140 (2020); O. Dmytruk, M. Schirò, PRB **103**, 075131 (2021)
- D. Loss, P. Goldbart, PRB **43**, 13762 (1991); J. Hauschild, F. Pollmann, SciPost Phys. Lect. Notes 5 (2018)